

Sensitivity study: $B^0 \rightarrow K^{*0}(K\pi^-) l^+l^-$ with a $J/\psi K^{*0}$ peaking background

K. Yoshihara (University of Hawai'i at Mānoa)
analysis performed and written by an LLM agent (Claude) under human review

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This study uses only publicly available tools and publications. No Belle II internal materials — data, MC, internal notes, or the software framework — are used. Everything is open-source and runs on a laptop.

1. Introduction

$B^0 \rightarrow K^0 l^+l^-$ ($l = e, \mu$) is a flavor-changing-neutral-current $b \rightarrow s l^+l^-$ transition, forbidden at tree level in the Standard Model and sensitive to new heavy particles in the loop. Belle first measured its branching fraction and forward-backward asymmetry using the beam-energy-constrained mass M_{bc} , the energy difference ΔE , and a $K\pi$ invariant-mass window around the $K(892)$, with asymmetric charmonium (J/ψ , $\psi(2S)$) vetoes in the dilepton mass that are wider on the low side for electrons because of bremsstrahlung (Wei et al. [Belle], PRL 103, 171801 (2009)). The vetoed $J/\psi K^*$ sample itself served as the control channel for efficiencies and PID. The most recent Belle II measurement of this same final state (arXiv:2206.05946) uses the identical strategy and quotes explicit numerical veto windows, J/ψ excluded at $M(\mu^+\mu^-)$ not in $[2.946, 3.176]$ GeV and $M(e^+e^-)$ not in $[2.846, 3.176]$ GeV, which this note adopts as the starting point and validates on its own simulation.

This note reproduces that selection logic on a fast-simulation sample to estimate the signal yield and statistical precision achievable with 1 ab^{-1} , and to quantify the size and control-region purity of the residual $J/\psi K^{*0}$ peaking background after the veto. Continuum and generic- $B\bar{B}$ combinatorial background are explicitly not modeled in this study (no such samples were generated); this is stated wherever it matters and discussed in Section 6.

2. Data samples and analysis tools

Samples were not regenerated for this study; two existing ntuples (produced by `make_ntuple_exclusive.py`, one row per reconstructed B candidate) were used directly.

Sample	Process	Generator model	Events generated	Flavor split	Seeds
sig_kstll.root	$B^0 \rightarrow K^{*0}(K\pi^-) l^+l^-$	EvtGen BTOSLLBALL	10000	50% e^+e^- , 50% $\mu^+\mu^-$	32 / 33

Sample	Process	Generator model	Events generated	Flavor split	Seeds
bkg_jpsikst.rB0t->	J/psi(->l+l-) K*0(K+pi-)	EvtGen (J/psi K* + J/psi->ll)	5000	50% e+e-, 50% mu+mu-	34 / 35

Relevant branches: `m_visible` (reconstructed $m(K\pi)$, i.e. the K^* candidate mass), `mbc`, `delta_e`, `m_ll` (reconstructed dilepton mass), `lep_flavor` (11 = e, 13 = mu). These ntuples carry no `true_mode` branch (unlike the `make_ntuple` output used in other notes of this framework): truth information here is simply which file a candidate came from.

The events were passed through the framework's fast detector simulation (track/cluster smearing, lepton and hadron PID) at ntuple-production time; only ~44-54% of generated events per flavor yield a reconstructed candidate row in the ntuple (Section 4, "reconstructed" cutflow stage) — this already includes basic reconstruction/acceptance losses.

Normalization to 1 ab^{-1} . The number of available B^0/B^0_{bar} mesons is

$$N(B^0 \text{ avail}) = 2 \times f_{00} \times \sigma(B\bar{B}) \times L = 2 \times 0.486 \times 1.1 \text{ nb} \times 1e9 \text{ nb}^{-1} = 1.0692e9$$

using $f_{00} = 0.486$, $\sigma(B\bar{B}) = 1.1 \text{ nb}$, and $L = 1 \text{ ab}^{-1} = 1e9 \text{ nb}^{-1}$. Per-event luminosity weights (expected events at 1 ab^{-1} per generated MC event, by flavor) are

$$w = N(B^0 \text{ avail}) \times BF(\text{mode}) \times BF(K^*0 \rightarrow K\pi) / N_{\text{generated}}(\text{flavor})$$

with $BF(K^0 ee) = 1.03e-6$, $BF(K^0 \mu\mu) = 1.05e-6$, $BF(K^0 \rightarrow K\pi) = 2/3$, $BF(J/\psi K^0) = 1.27e-3$, $BF(J/\psi \rightarrow ll) = 0.0597$ (same for e and mu), and $N_{\text{generated}} = 5000$ per signal flavor, 2500 per background flavor:

	w (events / 1 ab^{-1} per generated MC event)
signal, e+e-	0.1468
signal, mu+mu-	0.1497
J/psi K^*0 , e+e-	21.618
J/psi K^*0 , mu+mu-	21.618

The huge weight of the $J/\psi K^0$ sample reflects $BF(J/\psi K^0)$ being ~1000x larger than the signal BF : the charmonium veto (Section 3.4) must reject this background very efficiently.

All figures were produced with `plot_stacked` (luminosity-weighted, truth-split by sample and, where relevant, by reconstructed lepton flavor via a `lep_flavor` selection). All cut values were fixed with the `scan_cut` tool, scanning one variable at a time on top of the preceding cuts, using unweighted MC counts as the figure of merit $S/\sqrt{S+B}$ (a shape-level proxy; the weighted, physical yields are computed separately in Section 5).

3. Event selection

Four sequential cuts are applied, in the same order and with the same physics motivation as Wei et al. [Belle] and the Belle II $K^*(892)ll$ measurement.

3.1 K^* mass window (m_{visible})

Targets: wrong-mass K-pi combinations under the $K(892)$ lineshape (natural width 47 MeV) folded with detector resolution. In this simplified simulation both the signal and the $J/\psi K0$ background contain a genuine $K^*(892)$, so this cut is a resolution/acceptance cut rather than a signal/background discriminant — as confirmed by the scan, where the FOM falls monotonically as the window is tightened for both samples equally (scan_cut on m_{visible} , symmetric window):

window (GeV)	signal eff.	background eff.
± 0.05	74.8%	75.7%
± 0.075 (chosen)	83.6%	83.7%
± 0.10	87.8%	88.4%

Window chosen: $0.821 < m(K \pi) < 0.971$ GeV (0.896 ± 0.075 GeV), retaining $\sim 84\%$ of both samples while removing the far tails where a real combinatorial background (not modeled here) would dominate. Figure: plots/01_mvisible_presel.png (cut lines at the window edges).

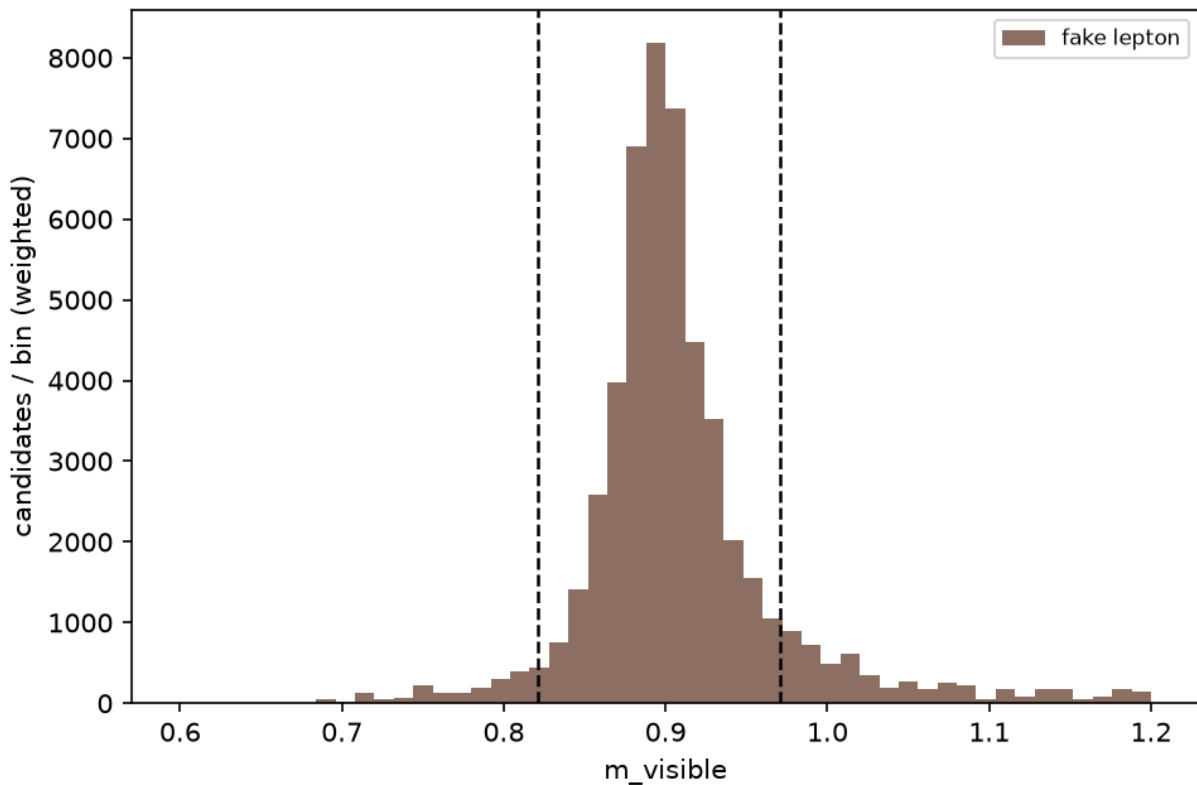


Figure 1: 01_mvisible_presel.png

3.2 Beam-energy-constrained mass (M_{bc})

Targets: combinatorial/continuum background populating low M_{bc} (not modeled here) and mis-reconstructed candidates; real signal peaks at the beam energy. Scan on top of the K^* window

(scan_cut, mbc, direction ">"):

Mbc > (GeV)	S	B	S/sqrt(S+B)
5.24	4103	1960	52.7
5.26	4048	1931	52.4
5.27 (chosen)	3999	1900	52.1
5.275	3953	1875	51.8
5.28	860	425	24.0

The sharp drop above 5.28 GeV marks the edge of the Mbc peak itself (beam-energy edge, ~ 5.29 GeV, smeared by detector resolution); cutting there would remove signal, not background. Mbc > 5.27 GeV is chosen, following Wei et al., retaining 96% of the K^* -window sample. Figure: plots/02_mbc_afterkst.png (cut line at 5.27 GeV).

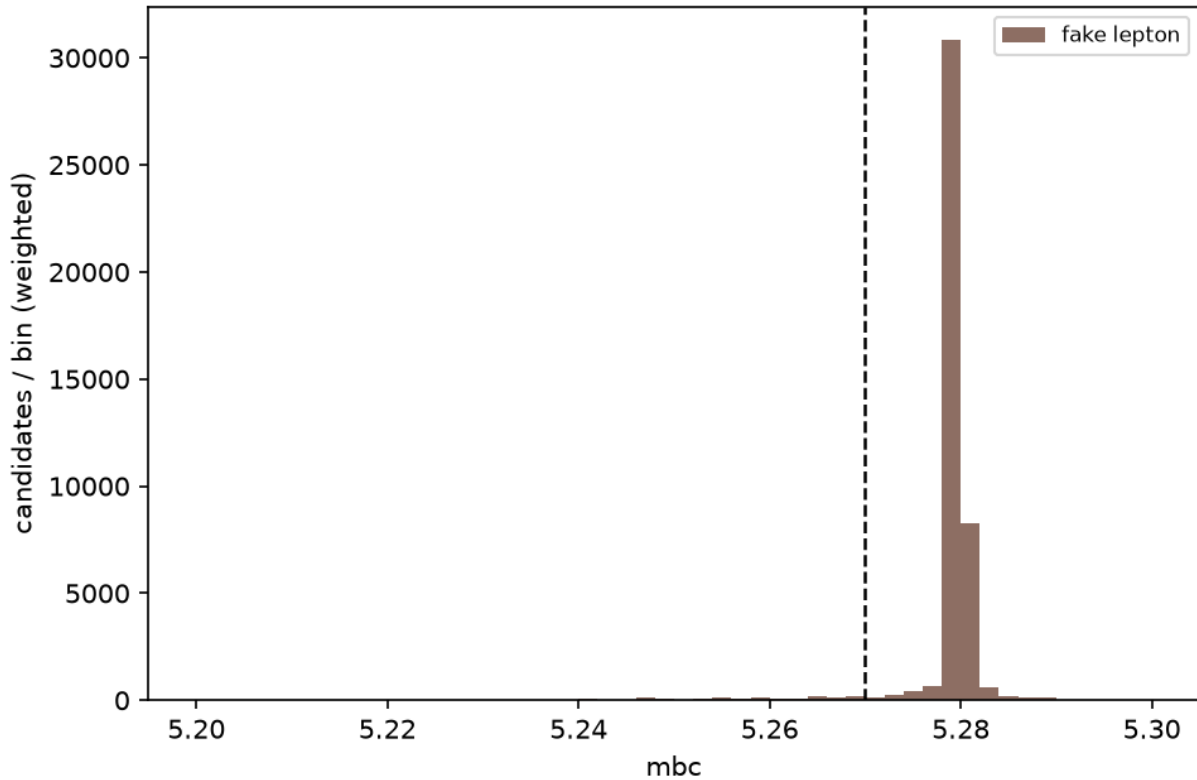


Figure 2: 02_mbc_afterkst.png

3.3 Energy difference (DeltaE), per lepton flavor

Targets: partially reconstructed or extra-energy background; for electrons the signal itself develops a low-side tail from final-state bremsstrahlung not fully recovered by clustering, motivating an asymmetric, wider low-side window (as in Wei et al.). Scans on top of the K^* +Mbc selection, separately per flavor (scan_cut, delta_e):

Electron channel (lower edge, direction “>”): FOM rises monotonically as the window is loosened and plateaus around -0.25 to -0.30 GeV (38.0-38.2), while the upper edge saturates at +0.05 GeV (S stops increasing beyond that). Chosen: $-0.25 < \Delta E < 0.05$ GeV.

Muon channel: the same scan shows a much smaller low-side tail (FOM 35.2-35.3 essentially flat from -0.10 to -0.30 GeV), consistent with the absence of bremsstrahlung; the upper edge again saturates at +0.05 GeV. Chosen: $-0.10 < \Delta E < 0.05$ GeV, a narrower low side than for electrons, reproducing the qualitative asymmetry reported by Wei et al.

Figures: plots/03_deltae_ee.png, plots/03_deltae_mumu.png (cut lines

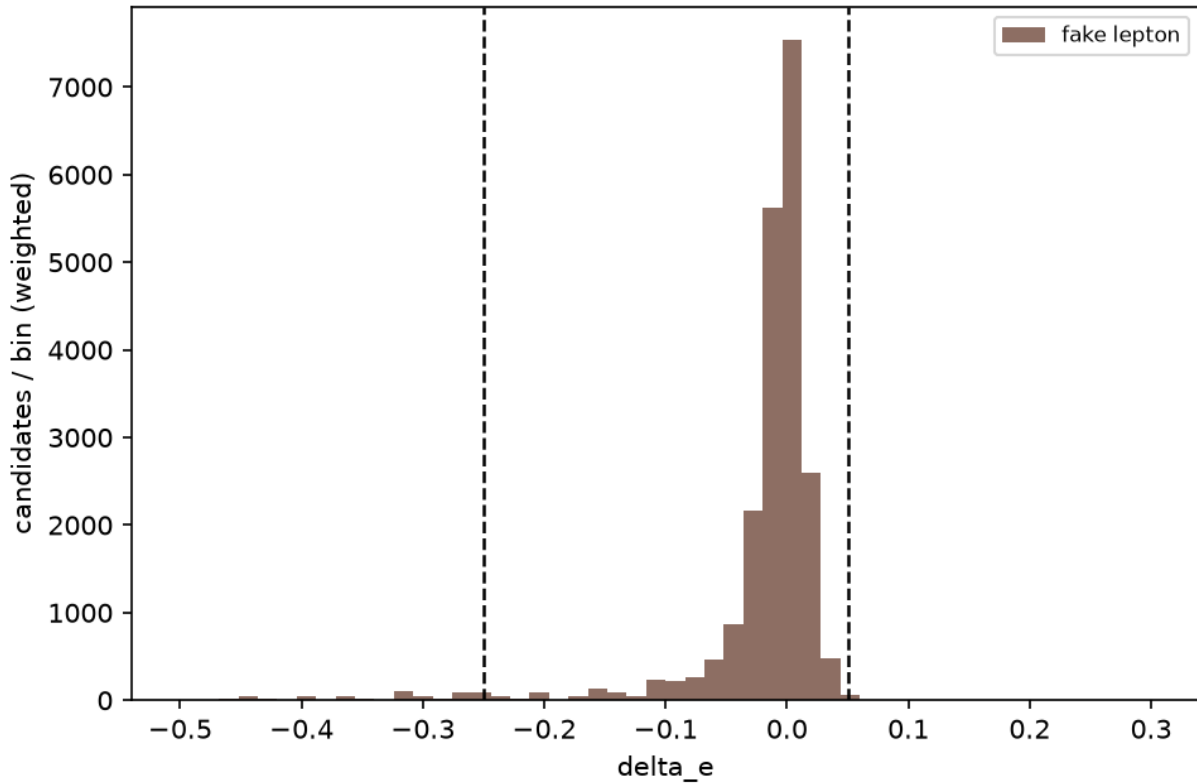
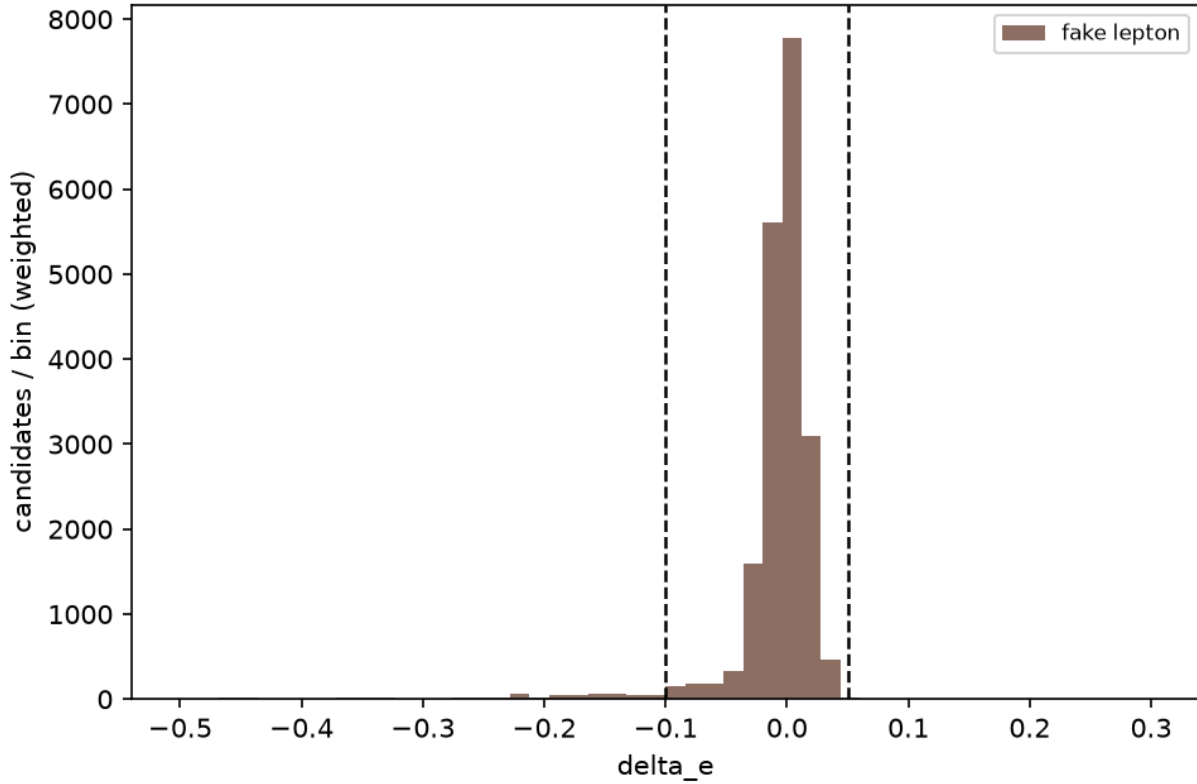


Figure 3: 03_deltae_ee.png



at the chosen edges).

3.4 Charmonium (J/ψ) veto in m_{ll} , asymmetric per flavor

Targets: $B^0 \rightarrow J/\psi(-\rightarrow l+l-) K^0$, the dominant peaking background, $\sim 1000\times$ the signal rate before any veto. The trade-off scan ($scan_cut$, m_{ll} , on top of $K+Mbc+\Delta E$, per flavor) brackets the peak from below and above:

Electron channel, lower edge (keep $m_{ll} < \text{threshold}$): background stays at 0-2 events up to 2.9 GeV then rises quickly (11 at 2.95, 30 at 3.0). Upper edge (keep $m_{ll} > \text{threshold}$): background jumps from 297 (>3.10 , inside the radiative tail region) to 4 (>3.13) to 0 (≥ 3.15).

Muon channel, lower edge: background is 0 up to 2.98 GeV, then rises (7 at 3.02, 26 at 3.05). Upper edge: background jumps from 316 (>3.10) to 1 (>3.13) to 0 (≥ 3.15).

Both flavors therefore have a “safe” plateau (background ~ 0 , signal near its local maximum) that brackets the Belle II window values; these are adopted as the final veto, confirmed rather than re-derived by the scan:

flavor	veto window (excluded)
e+e-	2.846 - 3.176 GeV
mu+mu-	2.946 - 3.176 GeV

Figures: `plots/04_mll_ee_veto.png`, `plots/04_mll_mumu_veto.png`.

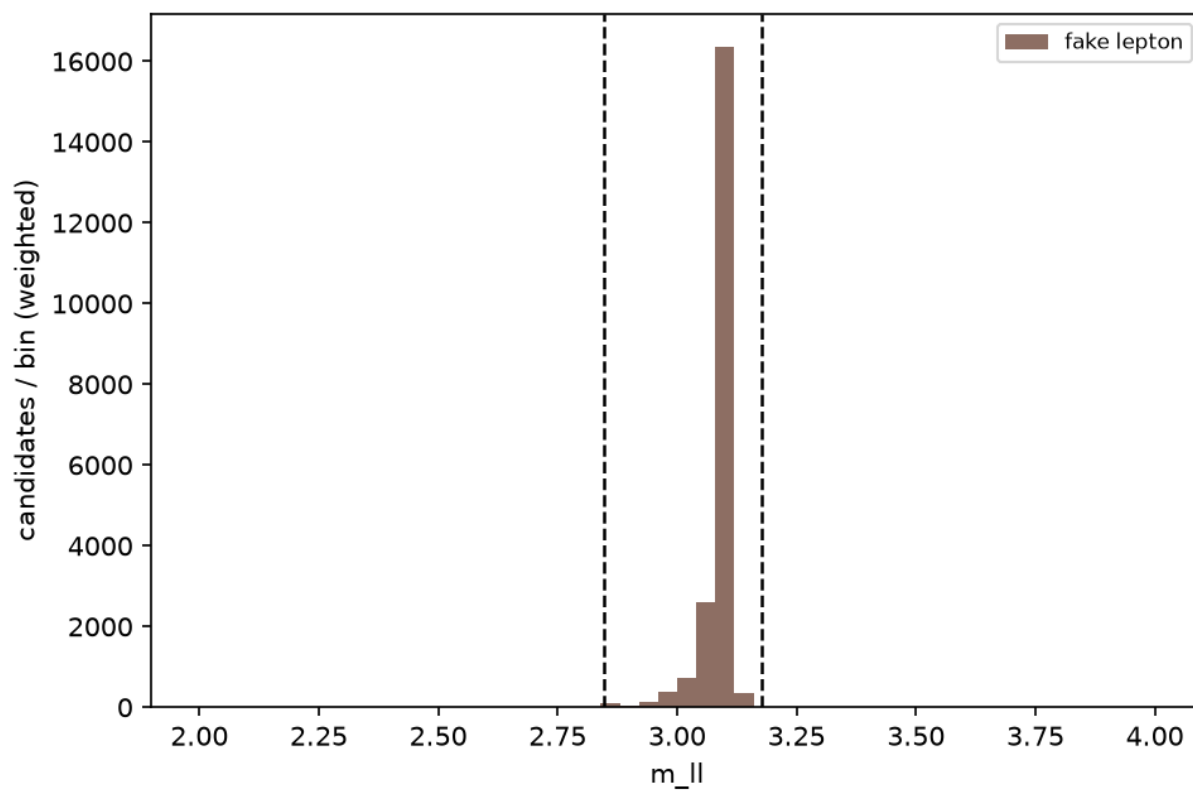


Figure 4: 04_mll_ee_veto.png

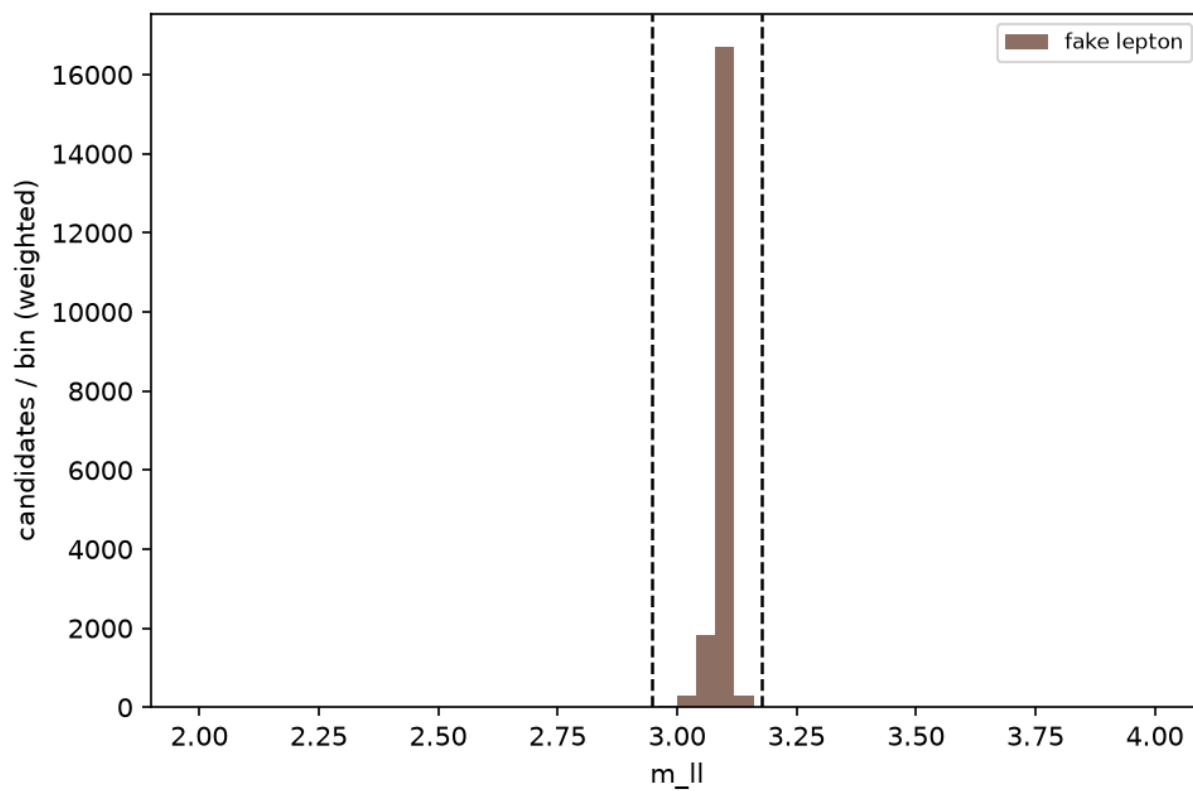


Figure 5: 04_mll_mumu_veto.png

4. Full cutflow (per flavor, per sample)

Efficiencies are cumulative relative to the number of generated events for that flavor (5000 for signal, 2500 for J/psi K*0 background); “reconstructed” is the fast-sim reconstruction/acceptance stage already built into the ntuple.

****Signal, B0 -> K*0 e+e-** (5000 generated)**

stage	N	step eff.	cum. eff.
generated	5000	-	100.0%
reconstructed	2701	54.0%	54.0%
K* window	2268	84.0%	45.4%
Mbc > 5.27	2138	94.3%	42.8%
DeltaE window	2104	98.4%	42.1%
J/psi veto	1946	92.5%	38.9 +- 0.7%

****Signal, B0 -> K*0 mu+mu-** (5000 generated)**

stage	N	step eff.	cum. eff.
generated	5000	-	100.0%
reconstructed	2292	45.8%	45.8%
K* window	1907	83.2%	38.1%
Mbc > 5.27	1861	97.6%	37.2%
DeltaE window	1819	97.7%	36.4%
J/psi veto	1682	92.5%	33.6 +- 0.7%

****Background, B0 -> J/psi(->e+e-) K*0** (2500 generated)**

stage	N	step eff.	cum. eff.
generated	2500	-	100.0%
reconstructed	1295	51.8%	51.8%
K* window	1068	82.5%	42.7%
Mbc > 5.27	986	92.3%	39.4%
DeltaE window	957	97.1%	38.3%
J/psi veto	2	0.2%	0.08 +- 0.06%

****Background, B0 -> J/psi(->mu+mu-) K*0** (2500 generated)**

stage	N	step eff.	cum. eff.
generated	2500	-	100.0%
reconstructed	1101	44.0%	44.0%
K* window	938	85.2%	37.5%
Mbc > 5.27	914	97.4%	36.6%

stage	N	step eff.	cum. eff.
DeltaE window	885	96.8%	35.4%
J/psi veto	0	0.0%	0.0% (0/2500)

The veto is the only cut that discriminates signal from background (step efficiency 92.5% for signal vs 0.2%/0% for the J/psi background); all preceding cuts have essentially equal efficiency on both samples, as expected since they share the same K^* and B kinematics.

5. Background estimation

The only background modeled in this study is the residual leakage of $B^0 \rightarrow J/\psi(-\rightarrow l+l-) K^0$ past the charmonium veto (Section 3.4); this is the dominant peaking background identified by Wei et al. and Belle II for this final state. Continuum $e^+e^- \rightarrow q\bar{q}$ and generic-BBbar combinatorial background are not modeled (no such MC was generated for this study) — see Discussion.

Control region. The J/psi veto window itself (before the veto is applied) is a natural control region for this background, exactly as used by Wei et al. and Belle II to calibrate efficiencies. Taking the candidates in that window after the $K^*+M_{bc}+\Delta E$ cuts and normalizing to 1 ab^{-1} :

flavor	J/psi K^0 yield in window	signal leakage in window	purity
e^+e^-	20645	23.2	99.89%
$\mu^+\mu^-$	19132	20.5	99.89%

The control region is essentially pure J/psi K^0 (>99.8%) in both flavors, confirming it can be used, as in the published analyses, to calibrate reconstruction/PID efficiency and the $M_{bc}/\Delta E$ shape of the peaking background directly from data rather than from simulation.

6. Results

Multiplying the final cutflow efficiencies (Section 4) by the total expected rate at 1 ab^{-1} (Section 2) gives the expected yields:

flavor	S (signal)	B (residual J/psi K^0)	S/sqrt(S+B)	relative stat. unc. sqrt(S+B)/S
e^+e^-	285.7	43.2	15.7	6.3%
$\mu^+\mu^-$	251.8	0.0 (0/2500 MC; 68% CL UL ~ 24.6)	15.9 (25.2 UL)	6.3% (6.6% UL)
combined	537.5	43.2	22.3	4.5%

With zero background MC events surviving the veto in the muon channel, the point estimate $B_{\mu\mu} = 0$ is quoted together with a simple 68%-CL Poisson upper limit (1.14 raw events, scaled by the muon-channel weight) to avoid overstating precision from a single MC sample with finite

statistics; a production analysis would use a much larger $J/\psi K^0$ MC sample to pin this down (see Discussion).

At 1 ab^{-1} the study projects order-500 selected signal candidates split roughly evenly between the two lepton flavors, each with about 6% relative statistical precision on its own, and about 4.5% combined, neglecting the background not modeled here (continuum, generic $B\bar{B}$). This is comparable in scale to the $\sim 40\text{--}80$ signal candidates reported by Belle II with 189 fb^{-1} (arXiv:2206.05946), scaled up by the larger 1 ab^{-1} sample size.

7. Discussion

Limitations, ordered by expected impact on a real measurement:

1. Continuum ($e^+e^- \rightarrow q\bar{q}$) and generic $B\bar{B}$ combinatorial background are not modeled. In every published $B \rightarrow K\ell\ell$ analysis these are the leading backgrounds, suppressed with continuum-suppression variables (Fox-Wolfman R2, event-shape/BDT discriminants) and fit with an ARGUS shape in M_{bc} plus a linear shape in ΔE (Wei et al.; Belle II arXiv:2206.05946 uses a 2D unbinned M_{bc} - ΔE fit rather than pure cut & count for this reason). A real analysis would add $q\bar{q}$ and generic $B\bar{B}$ MC samples, retrain the $K/M_{bc}/\Delta E$ cuts jointly with a continuum suppression cut, and extract the yield from a fit rather than a simple counting experiment. This is the single largest omission and likely dominates the true precision at 1 ab^{-1} .
2. $\psi(2S)$ veto is not modeled (no $B^0 \rightarrow \psi(2S) K^0$ sample was generated), even though it is part of the standard published selection at a similar $M_{\ell\ell}$ offset from its resonance. Its expected rate is smaller than $J/\psi K^0$ ($\text{BF}(\psi(2S) \rightarrow \ell\ell)$ is smaller and $\text{BF}(B \rightarrow \psi(2S) K^*)$ is also smaller) but should be added for completeness.
3. **Only one $J/\psi K^0$ MC sample size (5000 events)** was available; the muon-channel veto efficiency is limited to “0/2500 survive”, i.e. a background estimate with essentially no MC statistics behind it. A production study needs a much larger charmonium-background MC sample (or the data control region itself, as in Section 5) to quote a reliable central value rather than an upper limit.
4. No systematic uncertainties were evaluated (tracking $\sim 0.3\%$ /track, lepton PID $\sim 1\text{--}2\%$ /lepton, K^*/kaon PID, $N_{B\bar{B}}/f_0 \sim 1\text{--}2\%$, sub-decay BFs, all quoted at the level given by Wei et al. and the Belle II analyses above). At the $\sim 5\text{--}7\%$ statistical precision found here, PID and tracking systematics (few %) would already be a relevant, not negligible, contribution and should be added in quadrature.
5. Electron bremsstrahlung is treated only implicitly through the fast-simulation smearing and the wider ΔE /veto windows; no explicit photon-recovery (adding back a bremsstrahlung photon within a cone of the electron track) is modeled, unlike in the published analyses. This likely means the electron-channel efficiency and $m_{\ell\ell}$ resolution near the J/ψ edge are optimistic relative to real data.
6. **The K^* mass window was optimized on samples that both contain a genuine $K^*(892)$ **, so it could not be tuned against combinatorial background as in a real analysis; its value ($0.896 \pm 0.075 \text{ GeV}$) is a reasonable resolution-driven choice but should be re-optimized once combinatorial background is modeled.

8. Conclusion

Following the selection strategy of Wei et al. [Belle], PRL 103, 171801 (2009) and the Belle II B $\rightarrow$ K(892)l+l- branching-fraction measurement (arXiv:2206.05946) — a K mass window, Mbc and DeltaE requirements, and asymmetric per-flavor J/psi vetoes in m_ll — this study projects, from existing signal and J/psi K0 background ntuples reweighted to 1 ab⁻¹: about 286 e+e- and 252 mu+mu- signal candidates, with a residual J/psi K0 peaking background of about 43 events in the electron channel and consistent with zero (upper limit ~25 events) in the muon channel. The resulting statistical precision is about 6% per flavor and 4.5% combined, neglecting continuum and generic-BBbar combinatorial background, which are not modeled in this study and would need dedicated MC samples, a continuum-suppression cut, and a 2D Mbc-DeltaE fit (as in the cited Belle and Belle II analyses) for a publication-grade result. The J/psi K*0 control region is confirmed to be >99.8% pure in both flavors, supporting its use, as in the published analyses, to calibrate efficiencies directly from data.

Figures

- plots/01_mvisible_presel.png — K pi mass, K* window motivation.
- plots/02_mbc_afterkst.png — Mbc after the K* window, Mbc>5.27 GeV cut.
- plots/03_deltae_ee.png, plots/03_deltae_mumu.png — DeltaE per flavor, asymmetric window motivation.
- plots/04_mll_ee_veto.png, plots/04_mll_mumu_veto.png — dilepton mass, J/psi veto per flavor.